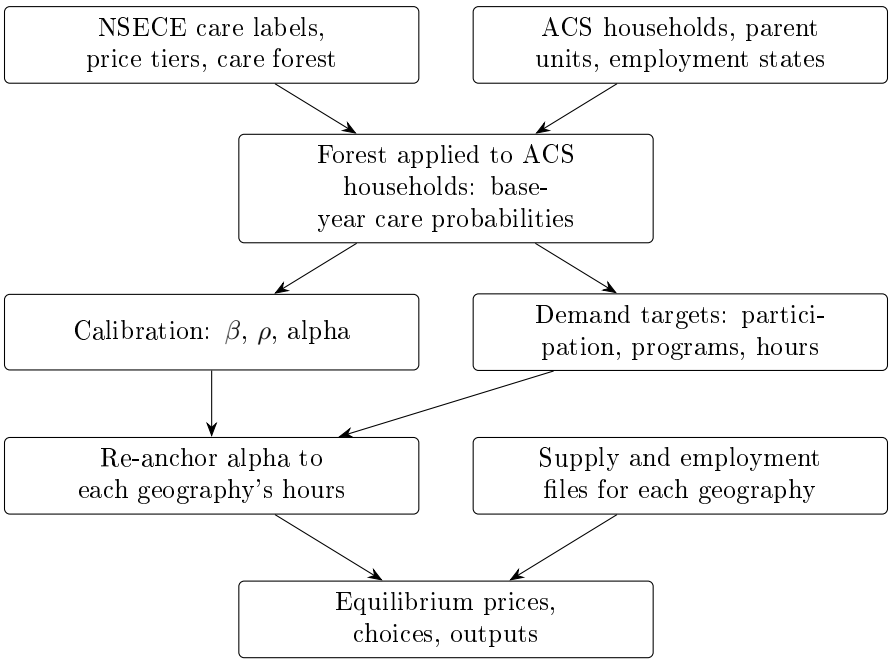
State Methodology (10/15)

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This memo follows the `state-inputs` branch from survey data to equilibrium prices. Processing and calibration run nationally for the 2019 base year. State simulations cover children ages 0–4 in 50 states, DC, and New York City, and New York City's separate run is excluded from national totals.

The national run is `main`'s. Italic lines identify what state runs change. The last section lists unresolved decisions.



1 Care model: NSECE

The care model learns from the 2019 National Survey of Early Care and Education (NSECE). The household survey gives each child's providers, hours, and payments. The center and home-based provider surveys give posted prices and staffing. The workforce survey gives wages and worker hours.

1.1 Care labels

Same as main, except that state runs classify household center payments with the posted-price threshold, not a fixed $4.75 an hour.

I count a provider when the household reports regular care and positive survey-week hours. A positive weekly payment means paid, and missing payments count as unpaid. I then assign each provider a label:

Label	Rule
K-8	School-age program (type 6).
Unpaid Center-Based	Center program (type 4) with no positive payment.
High- or Low-Priced Center-Based	Center program with a positive payment. The price distinctions below set the tier.
Paid or Unpaid Home-Based	Home care (types 1–3) outside the child's home, with no prior relationship to the family.
Paid or Unpaid Other	Everything else: care in the child's home, care by a relative or friend, and types 5, 7, and 8.

A child who uses several providers gets the first care type in this order:

K-8, high-price center, low-price center, unpaid center, paid home, other paid, unpaid home, other unpaid.

The child's hours are the summed hours in that type. A child with no counted provider is Parent Only. Fewer than 20 hours a week is part-time, and 20 or more is full-time. Each child ends with one of 15 outcomes: seven care types by two hour bands, plus Parent Only. The care model discards K-8 children.

1.2 Income rank

Same as main.

The care forest uses an income percentile from 1 to 100, ranked across the country. I cut income at the 1st through 99th percentiles, using households with children under 5 and positive income, weighted by child weight. I build the cutpoints separately for NSECE and ACS. Before I cut, I add a random number between 0 and 1 dollar to each income, so households with equal income do not all fall on one cutpoint. National and state baseline runs use nominal income. A price-adjusted rank, which divides income by the local price level first, stays available as a sensitivity (`ECEC_INCOME_RANK=national_rpp`). The child-weighted mean absolute rank change is 1.9 points, and 3.5% of children move more than 5 points.

The sensitivity's regional price parity (RPP) is the 2019 BEA all-items index, with the US at 100. ACS households use their state's RPP. NSECE households use their Census region's RPP, the child-weighted mean of its states' RPPs. The New York City run uses the metro RPP, 114.9.

1.3 Price distinctions

State runs vary the threshold by region and urban density and apply it to household payments. The national run matches main.

A paid center is high-price at or above the posted-price threshold. I apply the same threshold to household payments. The tiers represent distinct care options classified by price, not observed quality. National thresholds vary by age. State thresholds also vary by region and urban density:

$$\begin{pmatrix}\text{threshold}\\ \text{(region, urban density, age band)}\end{pmatrix} = \begin{pmatrix}\text{regional median price,}\\ \text{high urban density, age band}\end{pmatrix} \times \frac{\begin{pmatrix}\text{national median price,}\\ \text{this urban density, age band}\end{pmatrix}}{\begin{pmatrix}\text{national median price,}\\ \text{high urban density, age band}\end{pmatrix}} \tag{1}$$

Each median weights a center by its survey weight times full-time-equivalent enrollment, over positive prices. The posted price is the weekly rate divided by 40 hours, and a household's price is its weekly payment divided by its weekly hours. The age bands are 0–1, 2–3, and 4–5. Urban density is the NSECE community urban density (`COMM_URBAN_DENSITY`): high (85% or more of the nearby population is urban), moderate (30% to under 85%), or rural (under 30%). A center uses its own level, and a family uses its household's level. A regional rural cell holds only 5–35 centers per age band, and a moderate cell 28–134. A cell's own median is too noisy, so I use the national ratio and assume it holds in every region.

1.4 Care forest

Same as main.

A random forest predicts a child's care type and hours band from the family's features: child age and sex, income rank, household size, region, parent ages and sex, and caregiver employment. I train separate forests by number of children (one, or two or more) and by number of parents (zero, one, or two). A family with three or more children keeps its two youngest, with weights scaled to keep the total. Two-child families train on the joint outcome of both children, and I split the joint probability into one probability per child.

In ACS the caregiver is the mother in a two-parent family, otherwise the parent. In NSECE it is the respondent parent. ACS uses usual weekly work hours, and NSECE uses the parent's reported weekly hours. Both set the employment band by a 35-hour cutoff: none for zero or missing hours, part-time under 35, and full-time otherwise. The care forest has no urban density feature.

Each forest has 500 trees and a minimum node size of 10, and every split considers all features. Each forest trains on 50,000 weighted draws of family records.

2 ACS households

The simulator's households come from the 2019 American Community Survey (ACS) through IPUMS, with group quarters removed. I keep the person, family, work, income, tax, and poverty-measure variables, and add state and county from a crosswalk file.

Same as main, plus the state and county columns.

1. Children are persons ages 0–12.
2. Parents come from the mother and father links on each child. A unit has one or two parents, each at least 18. A child with no parent link has no unit.
3. Tax units come from the corrected tax-unit crosswalk. A unit's adjusted gross income (AGI) sums the fragments that belong to the corrected unit.
4. Hourly earnings equal annual earnings divided by (weeks worked $\times$ usual weekly hours). Where a parent has none, a two-step selection model fills it from age, sex, race, ethnicity, education, marital status, and the number of children in each age band, with a random draw.

2.1 Employment states

Same as main.

Each parent unit appears three times, once for each employment state of the caregiver: none, part-time, and full-time. For a state the caregiver does not hold, hours and weeks are the weighted

means among caregivers with that employment status, and earnings equal hourly earnings times hours times weeks. The difference between those earnings and the caregiver's observed earnings moves the unit's AGI by the same amount. A separate employment forest, not the care forest, gives the base-year probability of each employment state from the caregiver's age, education, marital status, race, ethnicity, earnings, the partner's earnings and hours, the number of children, and sex.

2.2 Base-year care probabilities

Same as main.

I run the care forest on every ACS family once for each employment state. The result is p_0, the base-year probability of each care outcome given the caregiver's employment. With nominal ranks, New York City uses the national forest predictions without a second pass.

3 Calibration

Calibration turns p_0 into utility terms that reproduce it, and estimates how much families respond to cost and income.

3.1 Choices

Same as main.

A choice is one combination of the caregiver's employment (none, part-time, or full-time) and a care outcome for each child (15 outcomes). A family with one child has $3 \times 15 = 45$ choices. A family with two or more children keeps its two youngest and has $3 \times 15 \times 15 = 675$ choices. Annual hours are 500 for part-time (10 hours, 50 weeks), 2,000 for full-time (40 hours, 50 weeks), and zero for none.

3.2 Net income and care cost

Same code as main. The inputs differ upstream.

$$\text{net income} = \text{AGI} - \text{taxes} - \text{child care cost} \tag{2}$$

AGI and taxes depend on the caregiver's employment state. Taxes come from Tax-Simulator donor records matched on AGI percentile, with the donor's effective tax rate. The care cost of each child is:

$$\text{child care cost} = \text{price wedge} \times \text{supply price of the sector} \times \text{annual hours} \tag{3}$$

The four market sectors are unpaid center, low-price center, high-price center, and paid home. Unpaid center care has a supply price of zero. Other paid care uses the NSECE national mean household payment as its base price. Unpaid home, other unpaid, and Parent Only cost nothing.

3.3 Price wedge

State runs break ties in the AGI ranks on a fixed number for each family. The national run matches main.

The price wedge is a family's payment relative to its sector's price, among the NSECE households that pay for that care type. For each paid care type, a quantile regression forest trained on NSECE predicts the wedge from the income percentile, region, number of parents, and the mean age of the

children under 5. Each family takes one random percentile of its forest, shared across care types. The price forest trains on NSECE household-income ranks but uses parent-unit AGI ranks when applied to ACS. Both use nominal income ranks. The care forest uses household income throughout. I clip wedges to the range 0.1 to 5.

I retain main's AGI ranks when applying the price forest to ACS. Switching to household-income ranks changes state mean wedges by at most 0.04, but individual wedges have mean absolute changes of 0.27–0.43 across care types.

3.4 Utility and estimation

Same as main.

$$\text{utility of a choice} = \beta \times \frac{\text{net income}^{1-\rho}}{1-\rho} \tag{4}$$

Net income has a floor of 1. I estimate one β and one ρ, shared by both family types, so that the model matches two elasticities:

Elasticity	Target
Share of caregivers who work, with respect to care cost	−0.15
Caregiver work hours, with respect to income	−0.05

I search a grid of β and ρ, refine with Nelder–Mead, and require both elasticities within 1% of target. The estimate keeps ρ between 0 and 0.15. The current calibration gives $\beta = 0.0003276$ and $\rho = 0.0264$, and both elasticities hit their targets.

3.5 Alpha

Same as main.

Alpha makes the model reproduce base-year probabilities, and it stays fixed across years and policy scenarios:

$$\text{alpha} = \log(\text{base-year probability}) - \text{utility} \tag{5}$$

New York City uses the national alpha. Only its re-anchoring to the city's own demand targets is separate.

4 Demand targets for each geography

A state's households carry national preferences, so I re-anchor each geography to its own level of care. The target is annual hours by care type.

New, and used in state runs only. Main has no demand targets.

4.1 Children in regular care

$$\text{children in regular care} = \text{population 0–4} \times \text{NSCH rate} \times \text{five-hour factor} \tag{6}$$

Population is the Census 2019 county estimate for ages 0–4. The National Survey of Children's Health (NSCH) rate is the share of children with at least 10 weekly hours of non-parental care,

pooled over 2016–2019 within each state. New York City uses New York's rate because NSCH does not identify the city. I scale this rate to regular care, whose NSECE screener requires at least 5 hours, using one national factor:

$$\text{five-hour factor} = \frac{\text{NSECE children in regular care}}{\text{NSECE children in care 10 or more hours a week}} \tag{7}$$

The factor is 1.098, based on an unweighted sample of 2,814 NSECE children. For the 50 states and DC, the result is 11.6 million children in regular care, 59.2% of the 19.6 million children ages 0–4.

4.2 Unpaid center children

Unpaid Center-Based care takes each geography's own program counts:

1. Pre-K is the larger of public-school pre-K from the Common Core of Data and an all-settings count. The all-settings count is Table 2 of the NIEER State of Preschool yearbook for a state, and the city's own count for New York City.
2. Funded Head Start and Early Head Start slots convert to enrolled children with the Office of Head Start enrollment ratios from the 2018–19 Program Information Report.
3. I remove home-based and family child care options, children age 5 and older, and Head Start children that pre-K already counts. The Head Start overlap share is one number for all geographies, the slot-weighted mean of the NIEER shares. Only 18 geographies have a NIEER share of their own, so I keep one pooled share.

4.3 Other care types

$$\text{children in type} = \text{children not in unpaid center care} \times \text{share of type among the other five types} \tag{8}$$

Children not in unpaid center care equal children in regular care minus unpaid center children. The shares are the simulator's own p_0, summed over the geography's households and weighted by child weight. The part-time share of a type is the same sum split by hour band. Each geography's mix therefore follows its income, parent employment, family type, and child ages, under the national preferences.

4.4 Hours and re-anchoring

$$\text{annual hours of type} = \text{children in type} \times \left(\text{part-time share} \times 500 + \text{full-time share} \times 2{,}000\right) \tag{9}$$

Each geography's file has six rows: Center-Based, Unpaid Center-Based, Paid Home-Based, Other Unpaid, Other Paid, and Unpaid Home-Based. Low-price and high-price centers share one row.

At simulation start, I solve one constant δ for each care type, shared by every household in the geography, so that base-year hours match the file. Parent Only has $\delta = 0$. A household's alpha for a choice moves by the sum of δ over its children's care types. The two center tiers share one δ, so the model sets the split between them. The run stops if a type has a positive target and no base support, if a target is zero, or if the solution does not converge.

The stored alpha was built at national supply prices, but a state run prices care at the state's. Before re-anchoring, I therefore rebuild each geography's alpha at its own 2019 supply prices. At simulation start, I check that the rebuilt alpha reproduces the base-year probabilities and that sector hours at the geography's prices equal the re-anchored hours. The checks pass in all 52 geographies,

with gaps below 10^{-8}. Without the rebuild, the 2026 change in the paid-center share of children 0–4, from the base-year target, runs from −9.4 (DC) to +7.0 (NC) percentage points across the 52 geographies (mean absolute 3.4). It correlates at −0.96 with the gap between the state's and the national 2019 low-price supply price. With the rebuild, it runs from −0.8 to +2.2 (mean absolute 0.9). The 2027 shifts are similar.

5 Supply and employment for each geography

The national supply parameters come from the NSECE calibration. Each geography's supply file changes three inputs: wages, staffing, and the residual.

New state supply inputs.

5.1 Wages

$$\text{wage of group} = \text{national wage of group} \times \frac{\text{state wage index of group}}{\text{national wage index of group}} \tag{10}$$

The two groups are workers without and with a bachelor's degree, with national wages of $12.06 and $19.70 an hour. The wage index mixes May 2019 hourly medians from the BLS Occupational Employment Statistics for childcare workers, preschool teachers, and preschool and childcare administrators. The mix uses the national ACS share of each group's workers in each occupation. New York City uses the New York metro area.

5.2 Staffing

$$\text{staffing index} = \frac{\text{state required staff per child}}{\text{national average of required staff per child, states weighted by children 0–4}} \tag{11}$$

The rules come from the 2017 Child Care Licensing Study (ICPSR 37700). For centers, required staff per child is the average of 1 over the child-to-staff ratio at five ages (11 months, 18 months, 35 months, 3 years, and 4 years), weighted by NSECE 2019 center enrollment at those ages. For homes, it is 1 over the maximum number of children in a family home. States without a family-home licensing ratio use an index of 1. New York City centers use the June 2019 city Health Code, and Idaho's center ratios come from its statutory points rule. The center index scales the three center sectors, and the home index scales paid homes. Both scale each worker group equally. I multiply the national labor requirement per child-hour by the index. Classrooms have more staff than the limits require, but the licensing indexes fit posted prices better than observed staffing.

5.3 Residual

Main uses a national residual for each sector; I now vary it by geography.

A sector's residual is its hourly price minus labor cost per child-hour. The national values are 0.51 for low-price centers, 4.61 for high-price centers, and 2.83 for paid homes. I now scale each sector's national non-labor residual by the geography's 2019 BEA housing-services price parity divided by 100. New York City uses the metro value, and unpaid centers keep a zero residual. The factor runs from 0.55 to 1.70 across the 52 geographies. Housing prices serve as a proxy for geographic differences in non-labor costs.

I compare prices across 46 states, weighted equally, with the National Database of Childcare Prices (NDCP). The comparison excludes CO, DC, IN, NM, NV, and New York City. The housing parity

correlates 0.71 with the residual that NDCP prices imply (weekly price divided by 40, minus hourly labor cost). The correlation of the model's center price with the NDCP state price rises from 0.78 to 0.89. The mean model price is $217.0 per 40-hour week, against $196.6 in NDCP.

5.4 Base-year labor

State runs only. National runs read base-year labor from the national supply file.

$$\text{base-year labor} = \text{total over sectors of labor requirement per child-hour} \times \text{base-year hours} \quad (12)$$

The base-year hours are the re-anchored hours, so labor demand equals base-year labor in the base year by construction.

5.5 Employment path

The state ratio file is new. The national path matches main.

The projected employment path of a geography, by sex and age band, is:

$$\text{rate} = \text{national projected rate} \times \frac{\text{state ACS rate}}{\text{national ACS rate}} \quad (13)$$

The national rate is the Current Population Survey history, projected forward on CBO's long-term labor force participation projections. The ratio comes from ACS two-year pools, and years past 2024 use the 2024 ratio. This path is not the final target. The simulator aggregates the fine age bands to four caregiver groups by sex, takes the path's growth since 2019, and uses it to scale the geography's base-year employment rate, the weighted mean of the employment-forest probabilities in the geography's parent units.

6 Simulation

Same as main, plus a loop over geographies, each with its own alpha, supply file, employment path, and warm start.

For each year, the simulator ages the 2019 households with national factors: population growth by age and marital status, hourly wage growth for earnings, GDP per person for income, and the CPI for prices. It then runs each geography as its own market, with no movement of workers or families between geographies (DC, inside the Maryland and Virginia labor market, is the case to watch):

1. It keeps the geography's households and takes its re-anchored alpha, supply file, and employment path.
2. L-BFGS-B solves low-price center, high-price center, and paid-home prices between $0.01 and $100, starting from the prior year's prices. Unpaid-center prices remain zero.
3. It draws one Gumbel shock for each choice and each household, drawing each household 10 times with one tenth of its weight per draw. Each household takes the choice with the largest utility plus alpha plus shock. Baseline and policy runs use the same shocks.

Demand for each sector is the weighted sum of choice probabilities times sector hours. Supply links demand to price through labor:

$$\text{wage of group} = \text{base wage} \times \text{wage growth} \times \left(\frac{\text{labor demand}}{\text{base-year labor}} \right)^{1/\text{supply elasticity}} \tag{14}$$

$$\text{price gap of sector} = \text{price} - \text{labor cost per child-hour} - \text{residual} \times \text{CPI growth} \tag{15}$$

The supply elasticities are 4 for workers without a degree and 2 for workers with one. The solver minimizes the sum over sectors of the squared price gaps. A solve counts as converged when the solver reports success and every gap is within \$0.01 an hour. A geography whose solve fails has no baseline for that year. The simulator writes each geography's results under its own folder.

7 Runs and the state setting

State setup uses `ECEC_URBAN_DENSITY=1` for the price tiers. Nominal income ranks come from `ECEC_INCOME_RANK=national`. Processing, calibration, and the interface record the setting, and each later stage stops if its setting differs from the interface's. The demand targets, supply files, and employment files load only when a run names states.

State runs rebuild alpha at the geography's own supply prices by default. The setting is `ECEC_STATE_ALPHA_PRICE`, and the value `national` keeps the alpha at national prices. One experiment switch is off by default: `ECEC_STATE_AGING=state` multiplies each age's national population growth by the state's growth relative to the nation at that age, from Census estimates for 2019 to 2024 (years after 2024 reuse the 2024 ratio; New York City uses New York). The run metadata records which geographies used each setting.

8 Open items

This proposal is not yet adopted.

8.1 State aging factors

Every geography ages with national factors by default. With `ECEC_STATE_AGING=state`, the model's 2019 to 2024 change in children 0–4 is within 0.6 percentage points of the Census estimate in each state I checked (KY, NJ, NY, and VA). The gap is the model's national path against the Census national change, 0.945 against 0.951. In the five test geographies (KY, NJ, NY, NYC, and VA), children 0–4 in 2026 as a share of 2019 move from 0.87–0.90 to 0.83–0.92: KY rises 1.8 points, NJ 3.3, and VA falls 0.4, while NY falls 5.7 and NYC 5.5. The 2026 paid-center shift changes by at most 0.4 percentage points. I have not checked the Weldon Cooper 2026 under-5 rows against the 2024 ratio that later years reuse.

Suggestion: assess the 2026 results of state ratios against the Weldon Cooper projections. Keep national wage and price growth.